

Lab 6 — Impossible Equations

Friday, December 4

Last updated: October 8, 2026.

Date is on the [schedule](#). Same day: [Quiz 10](#) (Week 13 eigenvalues — **not** this lab), a short [Lecture 40](#) (SVD / capstone launch), then this lab.

Credit: Lab points are for effort in class, not for a Canvas file upload. Work the hub exercises in Octave or Python on a laptop. On lab day you get a **self-contained one-pager** with the exercises written out, space for answers and new syntax, and a blank for instructor initials after a short check-in. Preview the [one-pager](#); write on the hard copy handed out in class. Least squares / SVD are assessed mainly via this lab, the capstone, and/or the final — not Quiz 10.

Resources: Lachniet Ch. 2 (least squares) · [Syntax sheets](#) · [Resources](#) · Lectures [38–39](#).

What we are doing

Solve an inconsistent overdetermined system in the least-squares sense: compute $\hat{x}$, plot data vs fit, report residuals and RMS, check $A^T r \approx 0$, and decide whether the model is useful in context.

Learning objectives

1. Compute a least-squares fit with Octave or Python.
2. Plot observed and fitted values.
3. Interpret residuals and an RMS error in context.
4. Check orthogonality of the residual to the columns of A .
5. Judge whether a fitted model is useful, not just report coefficients.

Math connections

Inconsistent $Ax = b$ and $\min_x \|Ax - b\|$ ([Lecture 38](#)); normal equations, RMS, orthogonality, and $A^T A$ PSD/PD ([Lecture 39](#)).

Key syntax (this lab)

Task	Octave	Python
Tall A , column b	<code>A = [1 0; 1 1; 1 2]; b = [1; 2; 2]</code>	<code>A = np.array([[1.,0],[1,1],[1,2]]); b = np.array([1.,2,2])</code>
Least-squares $\hat{x}$	<code>xhat = A \ b</code>	<code>xhat = np.linalg.lstsq(A, b, rcond=None)[0]</code>
Fitted / residual Norm / RMS	<code>bhat = A*xhat; r = b - bhat norm(r); norm(r)/sqrt(rows(A))</code>	<code>bhat = A @ xhat; r = b - bhat np.linalg.norm(r); np.linalg.norm(r)/np.sqrt(A.shape[0])</code>
Orthogonality	<code>A'*r</code>	<code>A.T @ r</code>
Scatter + fit line	<code>plot(t, b, 'o', t, bhat, '-')</code>	<code>plt.plot(t, b, 'o', t, bhat, '-')</code>

Indexing reminder: Boyd x_1 is 1-based. Octave is 1-based; NumPy is 0-based.

Demo (optional — run once)

Monday's tiny inconsistent system:

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

Octave:

```
A = [1 0; 1 1; 1 2];
b = [1; 2; 2];
xhat = A \ b
r = b - A*xhat
A'*r           % near zero
```

Python:

```
import numpy as np
A = np.array([[1., 0.], [1., 1.], [1., 2.]])
b = np.array([1., 2., 2.])
xhat = np.linalg.lstsq(A, b, rcond=None)[0]
r = b - A @ xhat
print(xhat, r, A.T @ r)
```

Expected: $\hat{x} \approx (1.167, 0.5)$; $A^T r \approx (0, 0)$.

Exercises

Pick Octave or Python for the term. Use column vectors for b and $\hat{x}$.

E1. Impossible by hand, possible by least squares. Let

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

- (a) Argue briefly that $Ax = b$ has no exact solution (two equations already force a third that fails).
- (b) Compute $\hat{x}$ with $A \backslash b$ / `lstsq`.
- (c) Report $\hat{b} = A\hat{x}$, $r = b - \hat{b}$, and $\|r\|$.

Expected (rounded): $\hat{x} \approx (1.167, 0.500)$; $\|r\| \approx 0.408$.

E2. Real-ish line fit. Five evening study-hour totals $t = 1, \dots, 5$ (hours) and self-reported focus scores b (0–10):

$$b = (3.0, 4.5, 4.0, 6.5, 7.0).$$

Use the model $b \approx x_1 + x_2 t$ with

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \end{bmatrix}.$$

- (a) Compute $\hat{x}$.
- (b) Compute $\hat{b}$, r , and $\text{RMS} = \|r\|/\sqrt{5}$.
- (c) In one sentence: on a 0–10 focus scale, is this RMS small enough to call the line useful?

Expected: $\hat{x} = (2, 1)$ (exact from normal equations); $\text{RMS} = \sqrt{3/10} \approx 0.548$.

E3. Plot. Using the E2 data, plot observed b vs t as markers and fitted $\hat{b}$ as a line (same axes). Label axes. Does the line capture the overall trend despite the dip at $t = 3$?

Expected answer: yes — upward trend; $t = 3$ sits below the line (visible residual).

E4. Orthogonality. For your E2 fit, compute $A^T r$. How close to $(0, 0)$ is it? Why should it be near zero?

Expected: entries on the order of 10^{-15} (roundoff); residual $\perp C(A)$ at a least-squares solution.

E5. Usefulness, not just coefficients. A classmate reports only “ $\hat{x} = (2, 1)$ ” with no plot and no RMS.

(a) What two checks would you still demand?

(b) Suppose instead $\text{RMS} = 3.5$ on the same 0–10 scale. Useful linear model? Why or why not?

Expected: (a) plot / residual look + RMS (or $\|r\|$) in context; (b) not useful — typical error is huge relative to the score scale.

E6 (stretch). PSD check. Form $P = A^T A$ for the E2 matrix A . Verify $P^T = P$ and, for $x = (1, -1)$, that $x^T P x = \|Ax\|^2 \geq 0$. Are the columns of A independent? Is P positive definite?

Expected: yes independent (Vandermonde-style line model); P PD, so $\hat{x}$ unique.

One-pager

Preview the [Lab 6 ticket](#); write on the hard copy in class.